

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN PHYSIOGRAPHY

MODULE - 3

PHYSIOGRAPHIC DIVISIONS **SIGNIFICANCE OF HIMALAYAS**

LONGITUDINAL DIVISION OF HIMALAYAS



THE NORTH AND NORTHEASTERN MOUNTAINS

4

ARUNACHAL HIMALAYAS

- 1 These extend from the east of the Bhutan Himalayas up to the Diphu pass in the east.
- 2 The general direction of the mountain range is from southwest to northeast.
- 3 These ranges are dissected by fast-flowing rivers from the north to the south, forming deep gorges.
- 4 Brahmaputra flows through a deep gorge after crossing Namcha Barwa.
- 5 Some of the important rivers are the Kameng, the Subansiri, the Dihang, the Dibang and the Lohit.
- 6 These are perennial rivers having the highest hydro-electric power potential in the country.

BRAHMAPUTRA RIVER



THE NORTH AND NORTHEASTERN MOUNTAINS

4

ARUNACHAL HIMALAYAS



7 An important aspect of the Arunachal Himalayas is the numerous ethnic tribal community inhabiting in these areas.

8 Some of the prominent ones from west to east are the Monpa, Daffla, Abor, Mishmi, Nishi and the Nagas.

9 Most of these communities practise Jhumming.

10 It is also known as shifting or slash and burn cultivation.



HUTAN

ASSAM

NAGALAND

ARUNACHEL PRADESH

DAFLA HILLS

MIRI HILLS

ABOR HILLS

MISHMI HILLS

Learning Space

THE NORTH AND NORTHEASTERN MOUNTAINS

4

ARUNACHAL HIMALAYAS



Transportation linkages are a biggest hindrance in Himalayas

11 This region is rich in biodiversity which has been preserved by the indigenous communities.

12 Due to rugged topography, the inter-valley transportation linkages are nominal.

13 Hence, most of the interactions are carried through the duar region along the Arunachal-Assam border.

THE NORTH AND NORTHEASTERN MOUNTAINS

5

EASTERN HILLS AND MOUNTAINS

- 1 These are part of the Himalayan mountain system having their general alignment from the **north to the south direction**.



- 2 They are known by different local names. In the north, they are known as Patkai Bum, Naga Hills, the Manipur hills and in the south as Mizo or Lushai hills.
- 3 These are low hills, inhabited by numerous tribal groups practising Jhum cultivation.
- 4 Most of these ranges are separated from each other by numerous small rivers.
- 5 The Barak is an important river in Manipur and Mizoram.

THE NORTH AND NORTHEASTERN MOUNTAINS

5 EASTERN HILLS AND MOUNTAINS

- 6** The physiography of Manipur is unique by the presence of a large lake known as 'Loktak' lake at the centre, surrounded by mountains from all sides.



Loktak lake of Manipur

- 7** Mizoram which is also known as the 'Molassis basin' is made up of soft unconsolidated deposits.

- 8**
- Most of the rivers in Nagaland form the tributary of the Brahmaputra.
 - The Barak river rises from the Manipur hills, which in turn is the tributary of Meghna;
 - The rivers in the eastern part of Manipur are the tributaries of Chindwin, which in turn is a tributary of the Irrawady of Myanmar.

SIGNIFICANCE OF HIMALAYAS



Life of people intricately linked with Himalayas

- 1 The Himalayas comprise the most dominating geographical feature of India.
- 2 No other mountain range anywhere in world has affected the life of people and shaped the destiny of a nation as the Himalayas in respect of India.
- 3 The Himalayas are the body and soul of the India. In a very special measure, the Himalayas constitute India's national mountain system.
- 4 The following few points will bring out the significance of Himalayan mountains to India.

SIGNIFICANCE OF HIMALAYAS

1

CLIMATIC INFLUENCE



Himalayas intercept the monsoonal winds

1

The Himalayas play a very significant role in influencing the climate of India in virtue of their **high altitude, length and direction**.

2

They effectively **intercept the summer monsoons coming from the Bay of Bengal and Arabian sea** and cause precipitation in the form of rain or snow, besides they **prevent cold continental air masses of Central Asia from entering into India**.

SIGNIFICANCE OF HIMALAYAS

1

CLIMATIC INFLUENCE



Cold desert conditions might be prevalent if there had been no Himalayas

3

Had there been no Himalayas, the whole of India would have been a desert in the absence of precipitation and its winters would have been very severe under the influence of cold air masses coming from Central Asia.

4

According to the latest Meteorological Studies, the Himalayas are also responsible for splitting the jet stream into two branches and these in turn play an extremely important role in bringing monsoons to India.

*As seen in Module 1 of
Indian Climate*

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTOR NO:2

THE HIMALAYAN MOUNTAINS

Question- What will happen if there had been no Himalayas?



- 1) Himalayas act as an **effective physical barrier for moisture bearing southwest monsoon winds.**
- 2) Himalayas divide the Bay of Bengal branch of monsoon winds into two branches_ one branch flowing along the plain region towards northwest India and the other towards southeast Asia.
- 3) If the Himalayas were not present the monsoon winds would simply move in to China and most parts of the Indian sub-continent would not have received sufficient amounts of rainfall as of now.

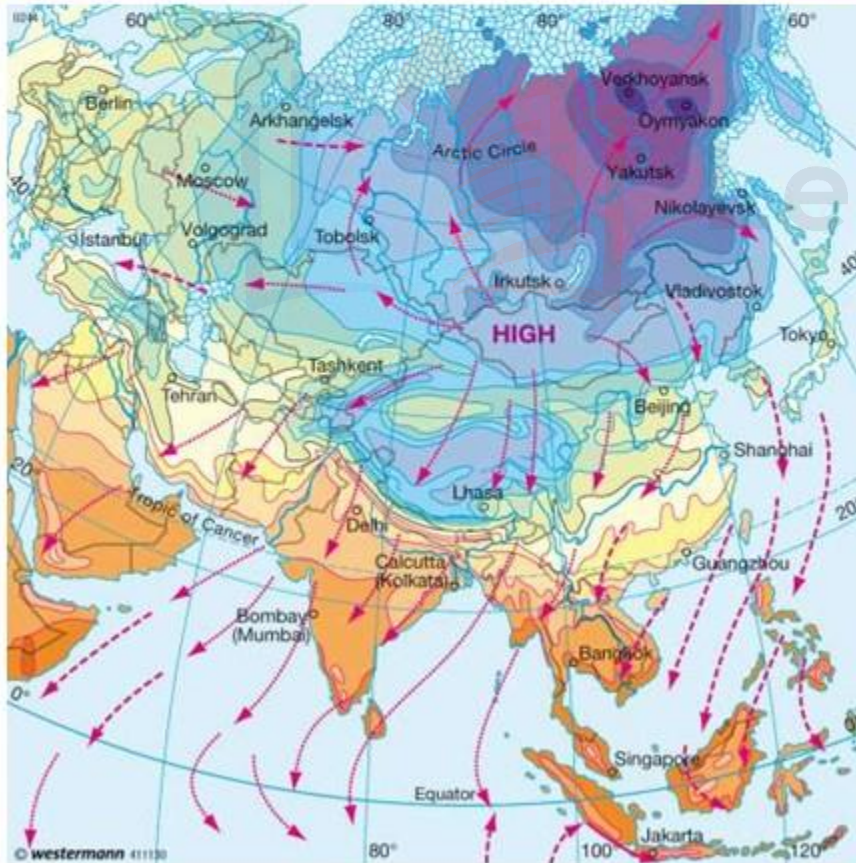
*As seen in Module 1 of
Indian Climate*

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTOR NO:2

THE HIMALAYAN MOUNTAINS

Question- What will happen if there had been no Himalayas?



- 4) Himalayas protect India from the cold and the dry air masses, which originates from Siberia and Central Asian region, this would have entered the Indian sub-continent owing to northeast trade winds thus making the place a cold dry desert with little or no vegetation.

*As seen in Module 1 of
Indian Climate*

FACTORS DETERMINING THE CLIMATE OF INDIA

FACTOR NO:2

THE HIMALAYAN MOUNTAINS

Question- What will happen if there had been no Himalayas?

- 5) In winters, the Himalayas bifurcate the *sub-tropical westerly jetstreams winds* and thus aid in the formation of winter-rainfall. Thus, Himalayas are responsible for fulfilling the water requirements of more than 1.3 billion population.
- 6) Without Himalayas, most of the place would have been a cold dry desert with little or no vegetation.
- 7) Also *the racial profile of India* would be very different as there would be very *strong Mongol attacks* because no natural defense would have been present and history might have been very different because of the extensive mongoloid invasion.

SIGNIFICANCE OF HIMALAYAS

2

DEFENCE



- 1 The Himalayas have been protecting India from outside invaders since the early times thus serving as a defence barrier.
- 2 But, the Chinese aggression on India in October, 1962 has reduced the defence significance of Himalayas to a considerable extent.
- 3 However, the defence significance of Himalayas cannot be ignored altogether.

SIGNIFICANCE OF HIMALAYAS

3

SOURCE OF RIVERS



- 1 Almost all the great rivers of India have their sources in Himalayan ranges.
- 2 Abundant rainfall and vast snow fields as well as large glaciers are the feeding grounds of mighty rivers of India.
- 3 Snow melt in summer provides water to these rivers even during dry season and thus these are perennial rivers.
- 4 The Himalayan rivers along with hundreds of their tributaries form the very basis of life in the whole of North India

SIGNIFICANCE OF HIMALAYAS

4

FERTILE SOIL



- 1 The great rivers and their tributaries carry enormous quantities of alluvium, while descending from the Himalayas.
- 2 This is deposited in the Great Plains of North India in the form of fertile soil making the plains one of the most fertile lands of the world
- 3 It has been estimated that the Ganga and Indus carry 19 and 10 lakhs tonnes of silt per day respectively and the silt carried by the Brahmaputra is even more.
- 4 It is, therefore, often said that the Great Plain of North India is a Gift of the Himalayas.

SIGNIFICANCE OF HIMALAYAS

5

HYDROELECTRICITY



- 1 The Himalayan region offers several sites which can be used for producing hydroelectricity.
- 2 There are natural waterfalls at certain places while dams can be constructed across rivers at some other places.
- 3 The vast power potential of the Himalayan Rivers still awaits proper utilisation.

SIGNIFICANCE OF HIMALAYAS

6

FOREST WEALTH



- 1 The Himalayan ranges are very rich in forest resources. In their altitude, the Himalayan ranges show a succession of vegetal cover from the tropical to the Alpine.
- 2 The Himalayan forests provide fuel wood and a large variety of raw materials for **forest based industries**.
- 3 Besides many **medicinal plants** grow in the Himalayan region.
- 4 Several patches are covered with grass offering rich pastures for grazing animals.

SIGNIFICANCE OF HIMALAYAS

7

AGRICULTURE



- 1 The Himalayas do not offer extensive flat lands for agriculture but some of the slopes are terraced for cultivation. Rice is the main crop on the terraced slopes.
- 2 The other crops in such slopes are wheat, maize, potatoes, etc.
- 3 Tea is a unique crop which can be grown on the hill slopes only.
- 4 A wide variety of fruits such as apples, pears, grapes, mulberry, walnut, cherries, peaches, apricot, etc. are also grown in the Himalayan region.

SIGNIFICANCE OF HIMALAYAS

8

TOURISM



Tourists in Himalayas

- 1 By virtue of their scenic beauty and healthy environment, the Himalayan ranges have developed a large number of tourist spots.
- 2 The hilly areas in the Himalayas offer cool and comfortable climate when the neighbouring plains are reeling under the scorching heat of the summer season.
- 3 Millions of tourists from different parts of the country as well as from abroad throng the Himalayan tourist centres to enjoy the natural beauty and to escape from the summer heat of the plains.

SIGNIFICANCE OF HIMALAYAS

8

TOURISM



Map shows the important tourist centres and hill resorts

4

The increasing popularity of winter sports and the craze to enjoy snowfall has increased the rush of tourists in winters also.

5

Srinagar, Dalhousie, Dharamshala, Shimla, Kulu, Manali, Mussoorie, Nainital, Almora, Darjeeling, are some of the important tourist centres in the Himalayas.

SIGNIFICANCE OF HIMALAYAS

9

PILGRIMAGE

*Amarnath cave**Vaishno devi temple*

1

Apart from places of tourist's interest, the Himalayas are proud of being studded with sanctified shrines, which are considered to be the abodes of the Gods.

2

Large number of pilgrims trek through difficult terrain to pay their reverence to these sacred shrines.

3

Kailas, Amarnath, Badrinath, Kedarnath, Vaishno Devi, Uttarkashi, Gangotri, Yamunotri, etc are important places of pilgrimage.

SIGNIFICANCE OF HIMALAYAS

10

MINERALS



- 1 The Himalayan region contains many valuable minerals.
- 2 There are vast potentialities of mineral oil in the tertiary rocks.
- 3 Coal is found in Kashmir.
- 4 Copper, lead, zinc, nickel, cobalt, antimony, tungsten, gold, silver, limestone, semi-precious and precious stones are known to occur at more than 100 localities in the Himalayas.

SIGNIFICANCE OF HIMALAYAS

10

MINERALS



5

Unfortunately, many of the mineral resources **cannot be exploited at the present level of technological advancement due to adverse geographical conditions.**

6

Further advancements in modern technology may help in exploiting these resources.

7

Also disturbing a fragile environment leads to **more pain than gain** and some of the hydroelectric power projects have already proved some of the disastrous effects of disturbing this fragile environment.

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